Self-Training-Based Unsupervised Domain Adaptation for Object Detection in Remote Sensing Imagery

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***Abstract*— We propose a novel two-stage cross-domain self-training (CDST) framework for unsupervised domain adaptive object detection in remote sensing. The first stage introduces the generative adversarial network (GAN)-based domain transfer strategy to preliminarily mitigate the domain shift for higher quality initial pseudo-labeled images, which utilizes the CycleGAN to transfer source-domain images to match the target domain. Moreover, the key issue in tailoring the self-training (ST) to unsupervised domain adaptive detection lies in the quality of pseudo-labeled images. To select high-quality pseudo-labeled images under the domain-shift circumstance, we propose hard example selection-based self-training (HES-ST) with the three key steps: 1) detector-based example division (DED), which divides the detected examples into easy examples and hard ones according to their confidence level; 2) confidence and relation joint score (CRJS)-based hard example selection, which combines two reliability levels calculated, respectively, by the detector and relation network (RN) module to mine reliable examples; and 3) union example (UE)-based training image selection, which combines both easy and reliable hard examples to choose target-domain images that may contain fewer detection errors. The experimental results on several remote sensing datasets demonstrate the effectiveness of our proposed framework. Compared with the baseline detector trained on the source dataset, our approach consistently improves the detection performance on the target dataset by 15.7%–16.8% mean average precision (mAP) and achieves the state-of-the-art (SOTA) results under various domain adaptation scenarios.**

***Index Terms*— Domain adaptation, object detection, remote sensing imagery, self-training (ST).**

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I. Introduction

RECENTLY, deep learning-based object detectors [1], [2], [3], [4], [5], [6] have achieved remarkable accuracy in computer vision and have been adopted in remote sensing. Best-performing detectors [7], [8], [9], [10], [11] are trained in a supervised manner. These fully supervised detectors (FSDs), due to their data-hungry nature, are heavily relied on a large number of samples with annotations. However, collecting annotations for remote sensing imagery is cost and difficult. To mitigate the demand for labels, one naive solution is to directly apply the detector trained by the labeled dataset to the new unlabeled dataset.

However, different large-scale remote sensing datasets generally have different distributions. For example, the xView [12] and the DOTA [13] datasets are captured by different satellites with varying locations, illumination, and weather conditions. Consequently, the detector would present poor performance on the new target data due to the data distribution discrepancy defined as the domain shift between two image domains. Unsupervised domain adaptation (UDA), which assumes labeled data as the source domain and unlabeled data as the target domain, provides an effective solution to solve this problem by transferring the knowledge from the source domain to the target domain. Domain adaptation for object detection aims to train a detector that can perform well on the target domain.

Over the past few years, UDA methods have exhibited impressive competence in classification [14], [15] and semantic segmentation [16], [17]. Compared to classification and semantic segmentation, object detection presents a more complex task since it involves predicting not only categories but also their locations. In the computer vision field, UDA for object detection has garnered significant attention from researchers and many methods have been proposed to tackle this problem, including generative, adversarial, and self-training (ST) methods. Generative training methods [18], [19], [20], [21], [22], [23], [24], [25], [26] typically generate domain-transferred images using a generative adversarial network (GAN) to directly align cross-domain data. Adversarial training methods [27], [28], [29], [30], [31], [32], [33], [34] attempt to learn domain-invariant features through an adversarial domain discriminator for cross-domain feature alignment. ST methods [35], [36], [37], [38], [39], [40], [41], [42] involve considering the predictions of a pretrained model to estimate pseudo-labels for unlabeled images. These

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pseudo-labeled images can be used for training to retrain the network in a supervised manner [35]. Ideally, the high-quality pseudo-labeled images should be accurate and complete, which enclose all objects of the images with accurate boxes, approximating the ground truth (GT). Generative training methods can work well for domain discrepancies caused by image-level/global style differences. However, they may not be well-adapted for instance-level/local feature differences. Adversarial training methods have been widely studied as the most popular strategy. However, these works are designed primarily for a two-stage object detector, e.g., Faster region-based convolutional neural network (R-CNN) [7]. This is because these methods highly rely on the region proposal mechanisms and the region of interest (ROI)-based local features of the two-stage detection architecture. They utilize these features to design local feature alignment strategies with respect to the foreground objects. This process of local alignment provides accurate adaptation results for object detection tasks. Unlike these methods, ST methods provide a more fine-grained way to allow the detector to learn instance-level features of target objects directly from pseudo-labeled boxes. Moreover, they can be applicable in any detector as it simply uses the predictions of the detector.

In this article, we investigate ST methods to address domain adaptation for object detection in remote sensing images. To the authors' best knowledge, only a few works [43], [44], [45], [46] have explored adversarial training methods for domain adaptive object detection in optical remote sensing images, while ST methods have not been well-investigated. To tailor the ST methods to UDA for object detection, the key issue is the quality of pseudo-labeled images. An ideal high-quality pseudo-labeled image means all the objects in this image should be accurately enclosed with pseudo-labeled boxes. It contains two requirements. One is the accuracy of detection boxes. If the proposed bounding box is misaligned or the category is incorrect, the learned features cannot well represent the objects. The other is the completeness of detections. If objects cannot be completely detected, the missed objects will be learned as background and detector will be confused. However, domain shift makes it difficult to generate high-quality pseudo-labeled images. As shown in Fig. 1, the baseline detector trained in the labeled source domain typically incurs two types of detection error when applied in the target domain: 1) false detections [false positives (FPs)], which reduce the accuracy of detections, as shown in Fig. 1(a) and (b). This may cause the detector to expend significant efforts attempting to learn misclassified objects or backgrounds (FPs) as foreground objects during the training procedure and 2) missed detections [false negatives (FNs)], which decrease the completeness of detected objects, as shown in Fig. 1(c) and (d). This may render the detector to regard foreground objects as backgrounds since the undetected objects are incorrectly used as the background to retrain the detector. Retraining with low-quality pseudo-labeled images can lead the baseline detector to accumulate errors and degrade detection performance.

To improve the quality of pseudo-labeled images, several works have been proposed using nature images. Inoue et al. [18] attempted to consider image-level annotations to select

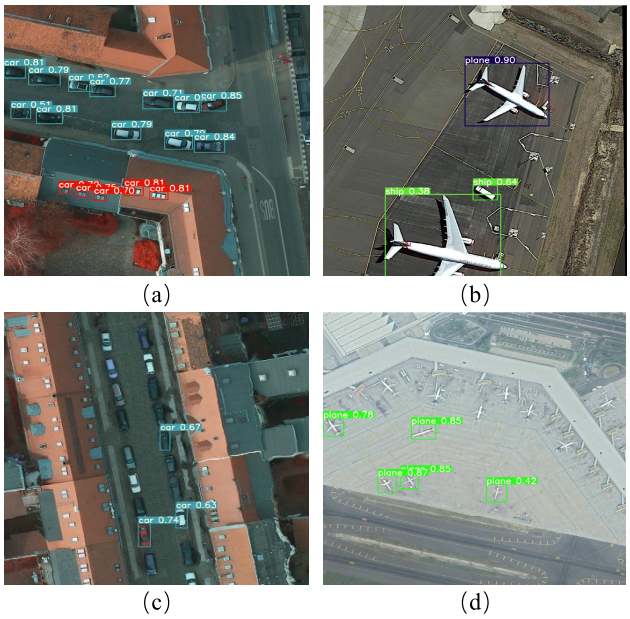


Fig. 1. Visualization of two types of detection errors. With the domain shift, a source-trained FSD may produce two types of detection errors when being applied to a target domain, making it difficult to generate high-quality pseudo-labeled images. (a) and (b) FPs, where backgrounds are wrongly detected as foregrounds or objects are assigned misaligned boxes or incorrect class labels. (c) and (d) FNs, where objects fail to be detected by the FSD, particularly in areas with dense objects.

the top confidence-ranked detections as high-quality pseudo-labels. However, it is not suitable for the UDA. Previous works use progressively confidence threshold [35] or confidence reweighting [47], [48] to omit unreliable noisy examples from the training process. However, these approaches can only select and keep a portion of examples for retraining since they do not sufficiently address the missed detection error. Recently, some methods [37], [38], [39] incorporated hard negative mining techniques [49] or emphasized difficult training examples [42]. It has been observed that training on examples the current model finds difficult can be more effective for the domain-adaptive detector. RoyChowdhury et al. [39] utilized the tracker [50] to exploit the temporal continuity between video frames to collect misdetected objects as pseudo-labels for augment. Although these strategies can enhance the quality of pseudo-labeled images with the most informative hard examples, they can be difficult to adapt to remote sensing images due to the heavy reliance on the temporal continuity typically found in video data. Moreover, they would involve retraining using significantly noisy examples when applied to remote sensing images due to the highly complex backgrounds and densely distributed objects with varying appearances in remote sensing imagery. In remote sensing community, ST methods for UDA in object detection have not been well-explored, especially for optical remote sensing images. To address the aforementioned key issue, we investigate a high-quality pseudo-labeled image selection method for unsupervised domain adaptive object detection in remote sensing.

In this article, we propose a cross-domain self-training (CDST) framework. For the pseudo-labeled image selection

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under the domain-shift circumstance, we proposed *hard example selection-based self-training* (HES-ST) with the following three key steps.

1) *Detector-Based Example Division (DED):* A generic detector is extended as a detection splitter to collect and divide easy and hard examples according to the detection confidence level.
2) *Confidence and Relation Joint Score (CRJS)-Based Hard Example Selection:* To maximize the number of reliable examples while minimizing the effect of FPs, the *relation network (RN)* module is introduced to classify the hard examples. Then, the CRJS is formed by merging two distinct reliability levels calculated, respectively, by the detector and RN, which serves as a criterion to automatically select reliable hard examples that contain foreground objects.
3) *Union Example (UE)-Based Training Image Selection:* To further reduce the effect of detection errors (FPs and FNs) and obtain high-quality training images from the target domain without any expensive cost, the *UE* strategy is proposed. This strategy combines easy and hard examples to choose target images that are less likely to be affected by domain shift as trainable images.

The proposed pseudo-labeled image selection method also depends on the initial detection performance, which is affected by data distribution discrepancy between domains. Therefore, to reduce the domain shifts, we introduce the *GAN-based domain transfer* (GDT) method as a preprocessing strategy. It uses CycleGAN [51] to generate target-like images by transferring the source image style to match with the target domain. A baseline detector that is trained by the original source images is then fine-tuned by using these generated images.

To quantitatively illustrate the effectiveness of the pseudo-labeled images in our method, we provide a detailed analysis of pseudo-label quality. Inspired by previous work on detection analysis [52] and evaluation measures [53], [54], we adapt several standard evaluation metrics to facilitate the quality analysis of pseudo-labeled images in ST for object detection.

The main contributions of this article are as follows.

1) We propose a two-stage CDST framework for unsupervised domain adaptive object detection in remote sensing; it can automatically adapt a detector to a new remote sensing image domain in an unsupervised manner.
2) We introduce the HES-ST with three key steps to choose high-quality pseudo-labeled images for the effective and stable ST process.
3) We adapt several standard object detection evaluation metrics to quantitatively analyze the quality of pseudo-labeled images; it can provide a comprehensive method for evaluation of the effectiveness of pseudo-labeled images for object detection.

The experimental results demonstrate that the proposed framework achieves state-of-the-art (SOTA) on several cross-domain detection tasks in remote sensing scenes under an UDA setting.

The rest of this article is organized as follows. Section II introduces the related works. Section III presents the proposed method. Section IV shows the experimental results. Conclusions are drawn in Section V.

II. Related Works

A. Object Detection

Deep learning-based object detection methods have been extensively studied in computer vision. On the one hand, region proposal-based methods, such as R-CNN method [1], Fast R-CNN [2], and Faster R-CNN [7], follow the traditional object detection pipeline with a two-stage process: generating region proposals and then classifying each proposal into different object categories. On the other hand, regression-based methods, such as SSD [3] and you only look once (YOLO) family [4], [5], [6], [10], [11], formulate the detection task as a one-stage regression: map directly from image pixels to position coordinates and class probabilities, without requiring a separate region proposal stage. In remote sensing, many object detectors [55], [56], [57], [58], [59], [60], [61] have been proposed by following these detection pipelines, which achieved promising performance. Ding et al. [55] introduced a multiscale representation and various combinations of enhancement strategies to optimize the backbone of the two-stage Faster R-CNN framework. Zakria et al. [57] improved the one-stage detection model YOLOv4 [10] by employing the proposed setting of the nonmaximum suppression (NMS) threshold and anchor allocation schemes. Teng et al. [59] introduced a novel clip long short-term memory-based object detection network for remote sensing imagery, which integrates global context clues and local spatial contextual correlations. Hou et al. [61] proposed the asymmetric feature pyramid network, dynamic feature alignment module, and area-intersection over union (IoU) regression loss to enhance a one-stage detector for the detection of multiclass objects with arbitrary orientations in remote sensing images.

B. Domain Adaptation

Domain adaptation transfers the knowledge learned from a well-labeled source domain to a poorly labeled target domain by mitigating the distributional variations. Under the UDA setting, the source domain is assumed to have abundant labeled information while the target domain does not have any labels. Standard UDA methods manage to formalize optimization problems to minimize the discrepancy distance between domains in certain ways, e.g., maximum mean discrepancy (MMD) [62] and correlation alignment (CORAL) [63] metric. In addition, adversarial learning is also effective for UDA by using a discriminator to learn domain-invariant features. Ganin et al. [64] pioneered the integration of adversarial learning in domain adaptation and proposed domain adversarial neural network (DANN) for cross-domain feature alignment. UDA methods have been successfully applied in remote sensing [65], [66], [67], [68], [69]. Chen et al. [65] investigated an augmented associative learning-based domain adaptation network for hyperspectral remote sensing image classification, employing cycle consistency criterion to achieve both domain-invariant and discriminative features.

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Wei et al. [67] proposed multiple domain adaptation fusion and multiple base classifier fusion technique for remote sensing image classification based on a neighborhood consistency using a samplewise adaptive weighting strategy. Zhang et al. [69] proposed a novel stagewise domain adaptation model for road segmentation of remote sensing images, using GANs for interdomain adaptation and adversarial ST for intradomain adaptation.

C. Domain Adaptation for Object Detection

Recently, a growing body of work investigated domain adaptation for object detection, using different methods to obtain a well-trained detector and address the domain-shift problem caused by distribution discrepancy between source and target domains. Existing methods can belong to three categories: generative, adversarial, or ST methods.

1) Generative Training Methods: Generative training methods usually generate domain-transferred images to directly align the cross-domain data. Image-to-image translation networks, such as CycleGAN [51], [70] and unsupervised image-to-image translation (UNIT) [71], [72], are employed to convert source images into target-style images or vice versa. On the one hand, the detector trained with labeled source images can generally achieve better detection performance on the source-style target images. Some works consider this strategy as one of the steps in their multistep adaptation framework to achieve progressive adaptation [19], unbiased mean teacher [20], or augmented feature alignment network [21]. On the other hand, the generated target-style source images can receive annotations from the source domain; thus, a baseline detector can be further trained in a supervised fashion [18]. Being fed by both real-source and target-like images, the detector may implicitly achieve feature alignment between two different domains. This approach has been utilized in several cross-domain scenarios, such as natural and cartoon domain [18], virtual and real-word domain [22], [23], daylight and nighttime domain [24], or RGB and thermal domain [25], [26].

2) Adversarial Training Methods: The main idea of adversarial training methods is to obtain domain-invariant features on the image and instance level for a detector to minimize the domain discrepancy. Those methods train a detector in an adversarial manner with the help of a domain discriminator and gradient reverse layer [64]. Chen et al. [27] propose the first work to formulate and address the problem of domain adaptation for object detection and designed a domain-adaptive faster R-CNN (DA-Faster) with three adaptation components, i.e., image-level adaptation, instance-level adaptation, and consistency regularization for a domain invariant region proposal network (RPN). Following this work, many similar methods were proposed, such as SW-Faster [28], MA-Faster [29], SA-Faster [30], HTCN [31], SCDA [32], and EPMs [33], and IMDA-YOLO [34], and have achieved promising results.

3) ST Methods: ST methods typically using the generated pseudo-labels along with the corresponding target images to train or fine-tune the detector. It can adapt the source-domain model to the target domain in a classwise adaptation and label-level adaptation manner, as the missing categorical and locational information for objects on the target domain is provided by pseudo-labels. Many approaches [35], [36], [37], [38], [39], [40], [41], [42] focus on generating effective pseudo-labels based on the detection results from a source-trained model. Inoue et al. [18] utilized image-level labels to help select the top-1 confidence detections as pseudo-labels and implement a two-step adaptation framework for cross-domain weakly supervised detection. Yu et al. [35] initialized the pseudo-labels by an intermediate domain with smaller gaps from the target domain and proposed two refinement components, namely, imbalanced sampling and label sharpening, to further improve the quality of pseudo-labels. Khodabandeh et al. [37] achieved robust learning by introducing an additional classification model to refine the noisy detections and compute the overall reasonable annotations for target data. Kim et al. [38] omitted unreliable examples from the retraining phase by considering the support region-based reliable score (SRRS) based on IoU and confidence, and a weak negative mining strategy is also introduced for conservatively choosing low training-loss negative examples as augment. Some other methods pay attention to mining the most training-difficult examples for retraining the detector [39], [40], [41]. In different surveillance scenes, Mou et al. [41] relabeled medium-confidence detections from the target video to retrain and evolve a generic pedestrian detector to a domain-adaptive scene-specific detector.

In remote sensing, following the adversarial training pipeline, related methods [43], [44], [45], [46], [47] were proposed. Koga et al. [43] applied both adversarial domain adaptation and CORAL domain adaptation to a region-based vehicle detection framework and improved the generalized performance in the target domain. Xu et al. [45] introduced a feature-aligned domain adaptive (FADA) object detector that considers aligning not only the cross-domain foreground features but also the confusing background and foreground features. Zhu et al. [46] proposed a reconstructed feature alignment network (RFA-Net) for the detection of remote sensing scenes with multicategories, which includes sequential data augmentation, sparse feature reconstruction, and pseudo-label generation.

III. Proposed Method

Consider a source domain $D_s = \{(I_i^s, y_i^s)\}_{i=1}^{N_s}$ with N_s well-labeled remote sensing images, where y_i^s refers to the GT annotations of an image I_i^s, and a target domain $D_t = \{(I_i^t)\}_{i=1}^{N_t}$ with N_t unlabeled remote sensing images. The source domain and the target domain are assumed to have different data distributions with a domain gap. Thus, a source-trained detector cannot provide satisfactory performance when evaluated on target-domain images.

CDST Framework Overview: The proposed CDST framework for unsupervised domain adaptive remote sensing object detection contains the following two steps as shown in Fig. 2.

1) Due to the significant domain shift, the baseline detector suffers poor detection accuracy on the target domain. This can lead the ST process to fail, resulting in negative transfer. Therefore, we conduct GDT to obtain transferred target-style source-domain images, which match the target domain with smaller gaps. Then, the baseline

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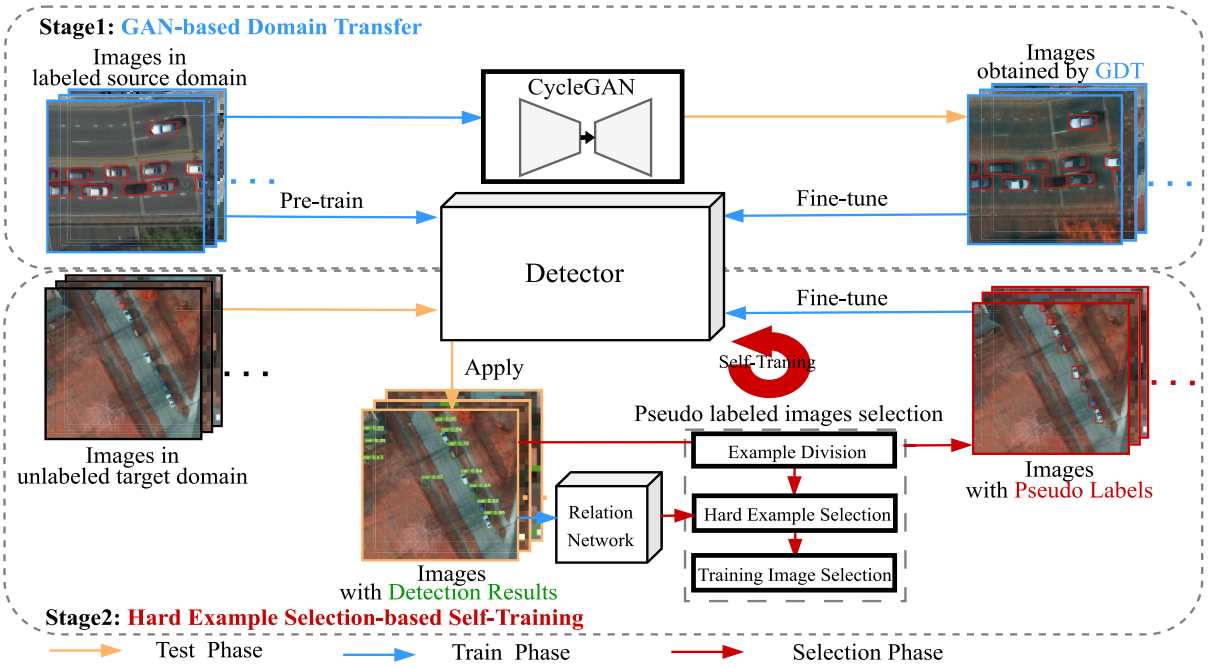


Fig. 2. CDST framework overview.

detector is fine-tuned by using such transferred images with source-domain annotations to reduce the inaccurate detections for the higher quality initial pseudo-labels. This step needs to be executed for only once to preliminarily bridge the domain gap and start the critical subsequent ST step.

2) We then perform HES-ST on the detector to allow it to learn better representations directly from the target domain for further adaptation. This is achieved by selecting pseudo-labels for the target-domain images from the detector's predictions, followed by fine-tuning the detector with these pseudo-labeled images. To prevent low-quality pseudo-labeled images from misleading the fine-tuning process, we present three key steps, which obtain reliable and complete pseudo-labels and select training images with fewer detection errors from the target domain. After each ST round, the detector might reduce the number of its uncertain detections (hard examples) and yield more accurate detections that can be used to obtain better pseudo-labels. Therefore, we could iterate the ST process multiple times until the detector stops boosting its performance.

A. GAN-Based Domain Transfer

Due to differences in the data source, imaging modalities, geographic location, weather and illumination conditions, etc., the discrepancies of appearance (e.g., texture, color, and brightness) are generally observed between the source-domain and target-domain remote sensing images. For example, the remote sensing RGB and color-infrared (CIR) images exhibit distinct image styles, as different detection bands can lead to highly dissimilar pixel value distributions between the two datasets [73].

We generate visually similar images to the target domain in order to capture such appearance differences. This process involves converting source-domain images and explicitly aligning their distributions with the target domain. Some traditional methods, such as graph matching [74] and histogram matching [75], focus on aligning the color distribution to perform simple mitigation. To align high-dimensional data effectively, we employ the UNIT network, CycleGAN [51]. The CycleGAN enforces a cycle-consistent mapping function learning process: an image from the source-domain data D_s is first translated to the target-domain data D_t by learning a mapping $G : D_s \rightarrow D_t$, then translated back with an inverse mapping $F : D_t \rightarrow D_s$, and the output should be identical to the original image. Once the mapping functions between D_s and D_t are trained, we convert source-domain images into the target domain to generate target-style transferred images. Meanwhile, the GT labels from the source domain can be directly replicated to the generated images, as the translation network is assumed to translate only the visual appearance between domains, with the position and pose of the object of interest remaining unchanged.

We first obtain the baseline detector by pretraining the detector with the labeled source-domain data. Then, we fine-tune the baseline detector on the target-style transferred images with annotations. This can bring several benefits to detector adaptation: 1) it can make the detector robust to appearance difference and implicitly achieve global feature alignment as both real-source and target-like images are fed into the detector and 2) it can preliminarily bridge the domain gap between the source domain and target domain $D_{s\rightarrow t}$ as the detector is trained on a similar target-domain distribution, making it more possible to perform ST for further domain adaptation in a stable manner. When the domain gap between $D_{s\rightarrow t}$ is too large, the source-domain-trained detector can predict many

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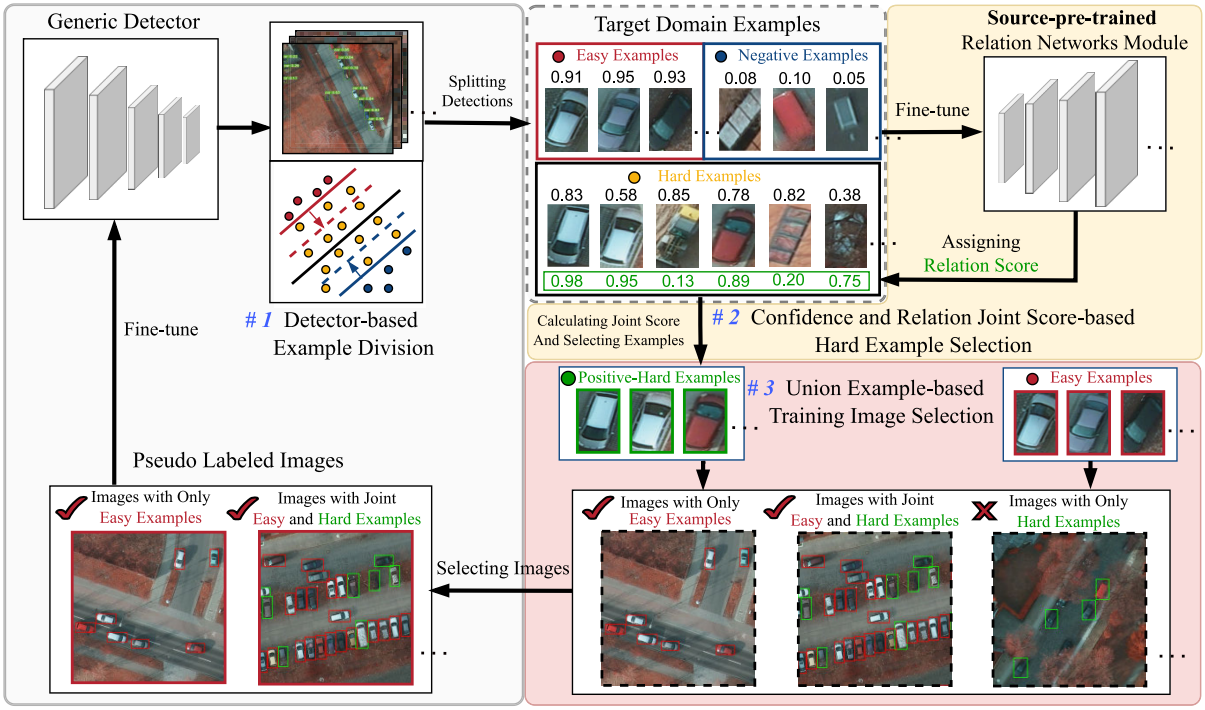


Fig. 3. Overview of our HES-ST approach.

inaccurate detection boxes, resulting in erroneous pseudo-labels. A smaller domain gap between $D_{s\to t}$ can mitigate such a problem, making the sequent ST less likely to fail. We only need to conduct this generate-and-finetune process once, and then, our ST can iteratively improve the quality of pseudo-labeled images, which are used to fine-tune the detector to improve the detection performance in the target domain.

B. Hard Example Selection-Based ST

The key to ST method in UDA for object detection is to select high-quality pseudo-labeled images to retrain the network according to its own predictions on the new target data. Unfortunately, inaccurate detection caused by FP detection error and incompletely detected objects due to FN detection error can both contribute to a decrease in the quality of pseudo-labeled images. Retraining with low-quality pseudo-labeled images can lead the detector to accumulate errors and degenerate detection performance.

Overview: To prevent low-quality pseudo-labeled images from overwhelming the ST process, we propose the HES-ST method and develop three key steps to address both FP and FN detection errors. Fig. 3 shows the overview of the proposed HES-ST. The YOLOv5 [11], which is used as the generic detector, is applied to the target domain to start the ST process.

First, the generic detector is extended to incorporate detection-based example division, where the detector is initialized by confidence-based positive and negative boundaries with a large margin. The detections are then collected and divided as easy, hard, and negative examples, respectively. We consider easy examples as reliable examples that contain foreground objects and further refine the indistinguishable hard examples in order to obtain a sufficient number of reliable examples for pseudo-labels.

Second, an *RN module* is introduced to classify the indistinguishable hard examples and provide the relation score as a reasonable measure of reliability in a dynamical training manner. Then, we combine two reliability levels calculated, respectively, by the detector and RN to form a joint selection criterion called *CRJS*. Using this criterion, reliable hard examples, which are called positive-hard examples, can be automatically selected from the collected hard examples.

Third, the *UE-based training image selection* is proposed to select high-quality training image that contain fewer detection errors. We simply combine easy and positive-hard examples to obtain the corresponding images as the training images while excluding images that only contain hard examples, as such images may be severely affected by domain shift and contain false or missed detections.

Finally, all the training images selected from the target domain along with the pseudo-labels from selected reliable examples can constitute the pseudo-labeled images, and we fine-tune the detector on such data. As the detector may reduce uncertain detections and yield more accurate detections, we gradually reduce the margin between the positive and negative boundaries, improving the ability to explore the feature space of examples in the target domain and helping to generate consistent and informative training data. This process is repeated multiple times until convergence or the detector stops boosting its performance.

In the following, we sequentially describe the three key steps in detail: 1) the *DED*, which aims to collect hard examples; 2) the *CRJS-based hard example selection*, which is introduced to automatically select trustworthy hard examples from the target domain; and 3) the *UE-based training image selection*, which is an image-level selection strategy for high-quality pseudo-labels of trainable images.

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1) DED: We extend the generic detector to incorporate *detection-based example division*. Let $H^0_{\text{det}}(x)$ indicate the initial baseline detector, which is first applied to the unlabeled target images I^t. All detection examples $e_i (i \in 1, 2, \ldots, N)$, which are cropped from detected bounding box with predicted coordinates and classes, are collected and divided into easy examples S_{easy}, hard examples S_{hard}, and negative examples S_{neg} according to the confidence level calculated by the detector

$$\begin{cases} e_i \in S_{\text{easy}}, & \text{if } H^0_{\text{det}}(e_i) > \beta + \dfrac{\theta}{2} \\ e_i \in S_{\text{hard}}, & \text{if } \beta + \dfrac{\theta}{2} > H^0_{\text{det}}(e_i) > \beta - \dfrac{\theta}{2} \\ e_i \in S_{\text{neg}}, & \text{if } H^0_{\text{det}}(e_i) < \beta - \dfrac{\theta}{2} \end{cases} \tag{1}$$

where β represents the decision hyperplane, and $\beta + (\theta/2)$ and $\beta - (\theta/2)$ are the positive and negative boundaries, respectively, around β according to the margin distance parameter θ. In practice, we keep β as constant at 0.5 and reduce θ from large to small values during training. Intuitively, at the beginning of the training, the baseline detector has poor performance with inaccurate detection in the target scene; therefore, by setting θ to a large value, we can collect most of the detections located between two boundaries to act as the hard examples, and they require further selection for pseudo-labels. Additionally, in this initial stage, only a few detections can achieve a high confidence score above the positive boundary to serve as easy examples, and they can reasonably be regarded as reliable examples that contain foreground objects and directly used to retrain the detector. Furthermore, after each ST round, the detector may yield better performance on the target domain; thus, more easy examples can be collected and the number of hard examples can also be reduced. To this end, we gradually adjust θ to a smaller value. Ideally, when θ is reduced to 0, it indicates that the detector cannot find any difficult objects of the target domain and becomes a well-performed detector on the target domain. We will further elaborate this process in Algorithm 2.

Through the DED, we collect both easy and hard examples and introduce an easy–hard mechanism to provide the following two benefits: first, the mechanism is expected to help us capture as many reliable examples as possible by considering not only easy examples but also a portion of hard ones with reliability. It may help us to generate reliable and complete pseudo-labels. Second, this mechanism can gradually place extra emphasis on training difficult hard examples while simultaneously constraining the detector to maintain detection accuracy on easy examples [39] during the iterative ST process. It may allow us to achieve effective and robust ST.

2) CRJS-Based Hard Example Selection: Since the domain shift might still cause unreliable predictions when applying the detector to the target domain, in addition to the foreground objects, considerable FPs (e.g., background regions or objects from categories different than the predicted ones) can be contained as noisy in the collected hard examples, as shown in Fig. 4. To obtain the foreground objects as reliable examples, we introduce an additional *RN* module to classify each hard example by providing relation score, which is the similarity between each hard example and the foreground class. Moreover, we then combine both the detector and RN module to select hard examples that may contain foreground objects by calculating the *CRJS*. Since the detector and RN have different training data, network structures, and objective functions, we expect that combining such two diverse models to classify each hard example can be more reliable. Intuitively, having both models agree on a hard example should make it more likely to be a correctly classified foreground object.

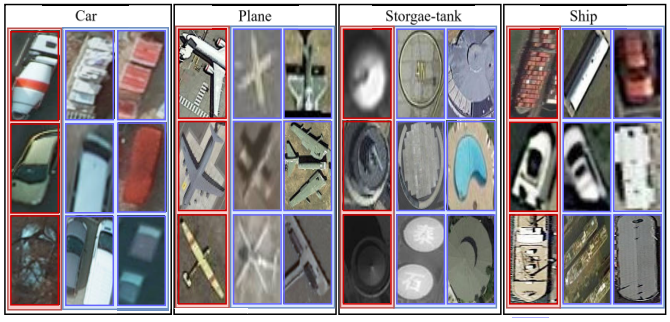


Fig. 4. Some collected hard examples, including foreground objects (in red boxes) and FPs (in blue boxes).

a) RN-based score: A standard RN [76] is a relation classifier CNN designed for image classification tasks. To differentiate between the foregrounds and backgrounds of hard examples, we employ the RN to classify each region image of hard examples, which is cropped from the predicted bounding box.

Algorithm 1 Union Example-Based Training Image Selection

Input: $E_t^{easy} = \{(e_i^{te})\}_{i=1}^{N_t}$, $E_t^{poshard} = \{(e_i^{tph})\}_{i=1}^{N_t}$ are easy example set and positive-hard example set, respectively, where e_i denotes an example that is cropped from the corresponding original image I_i^t of the unlabeled target domain $D_t = \{(I_i^t)\}_{i=1}^{N_t}$ without ground truth bounding box labels.

Output: $\hat{D}_t = \{\hat{I}_i^t, \hat{y}_i^t\}_{i=1}^{N_i}$ is the target domain training set, where $\hat{I}_i^t$ indicates the selected target images, $\hat{y}_i^t$ indicates the pseudo-labels that contain the bounding box coordinates and category information of selected examples.

Initialization:

1: Set $\hat{D}_t$ as an empty set

Selection:

1: Select the images $\hat{I}_i^t$ that contain easy examples e_i^{te} from the unlabeled target domain D_t

2: Update $\hat{D}_t$ by adding $\hat{I}_i^t$, $\hat{y}_i^t$, where $\hat{y}_i^t$ is obtained according to selected easy examples e_i^{te} and positive-hard example e_i^{tph}.

3: **return** $\hat{D}_t$

For each object class of the target domain, we need to train an RN model from scratch. Take the foreground–background binary classification for only one category of object as an instance, as shown in Fig. 5, the RN consists of two modules: an *embedding* module f_ϕ and a *relation* module g_φ, which are meta-learned end-to-end by a training example set data via episode-based training strategy [77], [78]. During each training iteration, one foreground class positive example with label $y_m^e = 1$ and one background class negative example with label $y_m^e = 0$ are sampled from the training example set to

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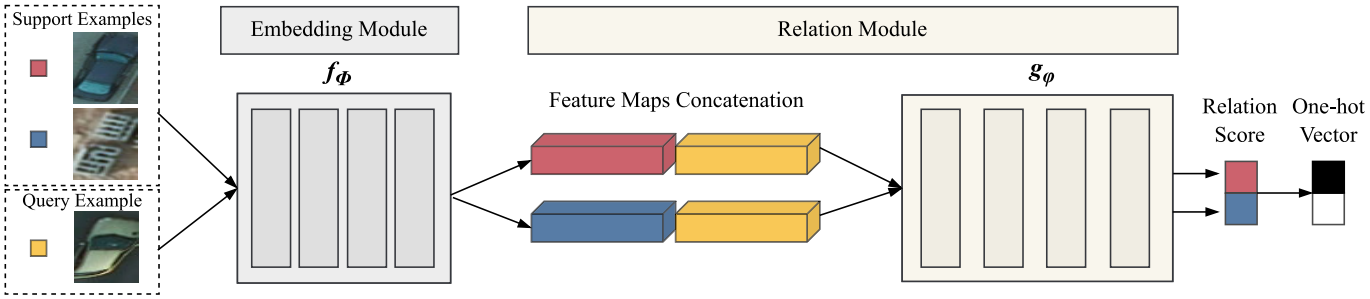


Fig. 5. RN framework for a two-way one-shot problem with one query example.

Algorithm 2 Hard Example Selection-Based Self-Training

Input: H_{det}^0 is the initial detector and some initialed parameters for the detector-based example division (DED) including β, θ. H_{rn}^0 is the initial RN model set on 2-way 1-shot. λ is a weight for calculating the CRJS, ν is a CRJS threshold to select positive-hard examples. r is the number of self-training rounds.

Output: H_{det} and H_{rn}

1: Training the initial detector H_{det}^0 on the source domain D_s
2: Training the initial RN H_{rn}^0 on source training example set E_s generated from the source domain D_s
3: Set E_t, E_t^{hard}, E_t^{easy}, E_t^{neg}, $E_t^{poshard}$ as an empty example set
4: $R = 0$
5: **while** ($R < r$ and $\theta > 0.3$)
6: applying H_{det}^R to detect object on unlabeled target domain D_t
7: collect easy, hard, and negative examples. update E_t^{easy}, E_t^{hard}, E_t^{neg} with easy, hard and negative examples, respectively. update $E_t = \{E_t^{easy}, E_t^{neg}\}$ with easy and negative examples.
8: $H_{rn}^{R+1} = Train(E_t)$
9: **for each** $e_k^{th} \in E_t^{hard}$ **do**
10: applying $H_{rn}^{R+1}(e_{i,j}^{th})$ and calculating final relation score by (4)
11: calculate $CRJS(e_k^{th})$ by (5)
12: **if** $CRJS(e_k^{th}) \geq \nu$ **then**
13: add e_k^{th} into the set of positive-hard examples $E_t^{poshard}$
14: **end if**
15: **end for**
16: using E_t^{easy} and $E_t^{poshard}$ to obtain the target domain training set $\hat{D}_t$ from the unlabeled target domain D_t by **Algorithm 1**
17: $H_{det}^{R+1} = Train(\hat{D}_t)$
18: $R = R + 1$
19: every round to update $\theta = \theta - 0.1$, $\nu = \nu - 0.05$
20: clear $E_t^{poshard}$ and $\hat{D}_t$
21: **end while**
22: **return** H_{det}, H_{rn}

constitute the *support* set $S = \{(e_m, y_m^e)\}_{m=1}^{N_m=2}$, as well as N_n labeled examples are randomly selected from a portion of the remaining two-class training examples to act as the query set $Q = \{(e_n, y_n^e)\}_{n=1}^{N_n}$. Since it is a two-class classification with one support example for each class, the task is described as the two-way one-shot problem. Examples e_m in the support set S and examples e_n in the query set Q are first fed into the embedding f_ϕ to extract feature maps $f_\phi(e_m)$ and $f_\phi(e_n)$. Then, the feature maps $f_\phi(e_m)$ and $f_\phi(e_n)$ are concatenated in depth by operator $\mathcal{C}(f_\phi(\mathrm{e}_m), f_\phi(e_n))$. The combined feature maps of the support and query are then fed into the relation module g_φ and finally output two relation scores $r_{m,n} \in [0, 1]$ for each query example e_n, which indicate the similarity between the query and the foreground class and background class of e_m, respectively. Finally, the relation score $r_{m,n}$ can be presented as follows:

$$r_{m,n} = g_\varphi\big(\mathcal{C}\big(f_\phi(\mathrm{e}_m), f_\phi(e_n)\big)\big), \quad m = 1, 2. \tag{2}$$

To train the RN model, the mean squared error (mse) loss is commonly used, following [76]. The RN loss function, as presented in the following equation, regresses the relation score $r_{m,n}$ to the GT, where matched pairs have similarity 1 and mismatched pairs have similarity 0:

$$L_{\mathrm{RN}} = \sum_{m=1}^{N_m}\sum_{n=1}^{N_n}\big(r_{m,n} - 1\big(y_m^e == y_n^e\big)\big)^2. \tag{3}$$

The RN model is trained in region images of training examples cropped from both source domain D_s and target domain D_t. Initially, the RN model, denoted as H_{rn}^0, is pre-trained by the source training example set E_s, where the foreground and background class examples are obtained by cropping from the source image based on the GT labels and randomly mining from the background regions, respectively. It may help the RN to extract transferable meta-knowledge from the source data, which can help to make predictions on the target data. Subsequently, the initial RN model H_{rn}^0 can be further fine-tuned using the updated target training example set $E_t = \{E_t^{\mathrm{easy}}, E_t^{\mathrm{neg}}$, which consists of easy examples with higher confidence and negative examples with lower confidence collected by the DED from the target domain. Assume there are a total of r rounds of ST during the entire HES-ST stage, this fine-tuning process is also performed r times. In each ST round, example division is first conducted, and the easy examples E_t^{easy}, negative examples E_t^{neg}, and hard examples E_t^{hard} are obtained. The RN model is trained using the updated training example set $E_t = \{E_t^{\mathrm{easy}}, E_t^{\mathrm{neg}}$ with easy and negative examples. One easy example and one negative example are randomly selected from the updated training example set E_t to form the support set, while N_n examples are sampled from a portion of the remaining training example set E_t to act as the query set. Then, the trained RN model can be used to calculate the final relation score for hard examples

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collected in this ST round (see the details in Algorithm 2). Since the training examples of E_t are updated along with the detector being updated and improved in each round, this would bring several benefits: first, as the detector is improved, more diverse and comprehensive examples can be collected to train the RN, improving its generalization ability to new target data. Second, as the DED is updated with novel and larger region of positive and negative examples by adjusting its boundaries over each round, the most up-to-date and relevant examples can be obtained to help RN to adapt to changes in the target domain and handle new types of hard examples that may arise.

Once the RN is trained, during each ST round, the unlabeled hard example set, $E_t^{\text{hard}} = \{(e_k^{\text{th}})\}_{k=1}^{N_{\text{th}}}$, collected by the DED from the target domain can become the query sets with N_{th} unlabeled hard examples, and the updated training example set E_t can provide the labeled foreground and background class examples to serve as a support set. Then, the episode tasks can be formed during the inference procedure, and we can easily calculate foreground class similarity r_{pos} and background class similarity r_{neg} for each hard example e_k^{th} as in (2). Moreover, since we are mainly interested in hard examples containing foreground objects, we apply a conservative restriction on the final relation score, denoted as $H_{\text{rn}}(e_k^{\text{th}})$, of each hard example e_k^{th}. Specifically, to be qualified as a foreground object and receive a relation score, a hard example should be more likely to belong to the foreground rather than the background. Otherwise, it will be assigned a score of 0 and considered as a background. Therefore, the final relation score $H_{\text{rn}}(e_k^{\text{th}})$ can be defined as follows:

$$H_{\text{rn}}\left(e_k^{\text{th}}\right) = \begin{cases} r_{\text{pos}}, & \text{if } r_{\text{pos}} > r_{\text{neg}} \text{ and } r_{\text{pos}} > 0.5. \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

b) Positive-hard examples selection: As mentioned before, a portion of hard examples that contain foreground objects can serve as positive-hard examples e_k^{tph} via selection. To obtain positive-hard examples, we combine both detector and RN to identify each hard example e_k^{th} in the target hard example set E_t^{hard}. Intuitively, having both models agree on a hard example should make it more likely to be a correctly classified foreground object. Specifically, for each hard example e_k^{th}, we combine the confidence $H_{\text{det}}(e_k^{\text{th}})$ from the detector and the relation score $H_{\text{rn}}(e_k^{\text{th}})$ from the RN module in a linear interpolation manner to calculate the *CRJS*, denoted as $H_{\text{CR}}(e_k^{\text{th}})$, and it can be defined as follows:

$$H_{\text{CR}}\left(e_k^{\text{th}}\right) = \lambda H_{\text{det}}\left(e_k^{\text{th}}\right) + (1-\lambda) H_{\text{rn}}\left(e_k^{\text{th}}\right) \tag{5}$$

where $\lambda \in [0, 1]$ is a tunable weight; when setting a high value of λ, it trusts the detector's detection confidence $H_{\text{det}}(e_k^{\text{th}})$ more than the RN's prediction $H_{\text{rn}}(e_k^{\text{th}})$. In a contrary setting, as more hard examples with low confidence can be selected for high relation score, it may help to generate more complete pseudo-labels, but also increase the risk of incorrect pseudo-labeling (PL). In this way, the CRJS $H_{\text{CR}}(e_k^{\text{th}})$ can be assigned to each hard example e_k^{th} as a criterion for selection, and then, considerable reliable positive-hard examples e_k^{tph} can be obtained through an easily set threshold ν.

3) UE-Based Training Image Selection: After mitigating the false detection (FP) error and obtaining reliable examples that contain foreground objects, we can select the images where these objects are located as training images to retrain the detector. As shown in Fig. 3, based on the two types of reliable examples, namely, "easy" and "positive-hard," three types of target images can be obtained: images with only easy examples, images with joint easy and positive-hard examples, and images with only positive-hard examples.

Images With Only Easy Examples (Imgs_Easy) are target-domain images that contain only higher confidence detections after the testing phase, indicating that such images are less likely to be affected by the domain shift (i.e., having a more similar distribution with the source domain data), and the source trained detector is less likely to be confused or produce false/missed detections on such images from the target domain. Given this assumption, we believe that the Imgs_easy contain comparatively lower overall training risk and thus can serve as pseudo-labeled images for each ST round. They may constrain the detector to maintain its posteriors on object correctly detected initially (e.g., easy examples) and provide the detector with more real background features from the target domain.

Images With Joint Easy and Positive-Hard Examples (Imgs_Joint) are the highly desirable trainable images that may be almost unaffected by the false and missed detection errors. This is because the objects of easy examples have been detected and the hard ones provide further supplementation after filtering out the noise. As a result, ideally, each object on the Imgs_joint can be accurately pseudo-labeled. We argue that the Imgs_joint may along with higher quality pseudo-labels that are reliable and complete, approximating the GT labels. Therefore, the Imgs_joint are deemed to possess higher training value and should be given priority utilization for the purpose of retraining the detector. They can allow the detector to learn examples that are difficult to the current model and maintain detection accuracy on easier ones, resulting in an effective and stable ST process.

Images With Only Positive-Hard Examples (Imgs_Hard) are target-domain images that contain only hard examples the model is uncertain about and finds difficult, thereby suggesting that they may suffer severely from the domain shift. When applying in such target-domain images, the source-trained detector may be too confused to discriminate the objects from the background or fail to detect objects, leading to detection error, especially the FNs. Although selected examples of the Imgs_hard are informative, the quantitative incomplete of the object problem is often unavoidable due to the overwhelming presence of undetected objects in the Imgs_hard. Consequently, we contend that employing the Imgs_hard as training images can lead to a decrease in the overall quality of the pseudo-labeled images and poses a high risk in the training procedure. We will later demonstrate this problem of pseudo-labeled image quality in Section IV-D2.

Based on our observations, we exclude the Imgs_hard and choose the Imgs_easy and the Imgs_joint as the training images to retrain the detector in each ST round. Additionally, it is noted that the combination of Imgs_joint and Imgs_easy represents exactly all the images including easy examples. Therefore, in practice, we only need to select the images that contain easy examples from the target domain to serve as training images. The *UE-based training image selection*

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TABLE I
DATASET DESCRIPTIONS AND COMPARISON BETWEEN THE SOURCE AND TARGET DOMAINS

	Dataset	Class	Bands	Device	GSD	Size-I	#Images	#Objects
D_s	UCAS-AOD	Car	RGB	GoogleEarth	0.5 - 1.0 m	1280 × 659 - 1372 × 941	510	7114
D_t	ISPRS Potsdam	Car	IRRG	-	0.05 m	$6k \times 6k$	38	20711
D_s	xView	PE+SP+ST	RGB	WorldView-3	0.3 m	$2.5k \times 2.5k$ - $5k \times 5k$	426	15031
D_t	DOTA	PE+SP+ST	RGB	GoogleEarth+	0.1 - 4.5 m	800 × 800 - $4k \times 4k$	792	117036

• 'PE+SP+ST': Plane, Ship, Storage-tank
• 'GoogleEarth+': Google Earth, GF-2, JL-1

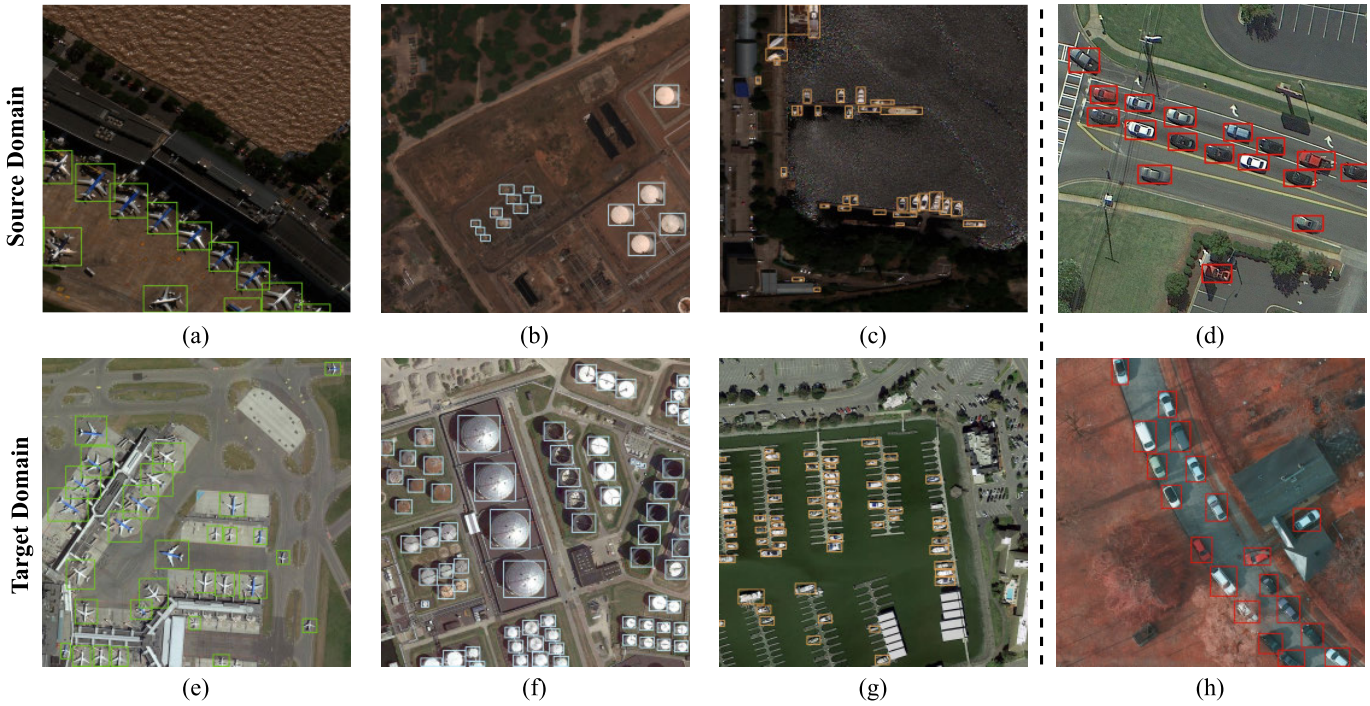


Fig. 6. Image comparison of different remote sensing image datasets for two adaptation scenarios. For cross-satellite adaptation from the xView dataset to the DOTA dataset. (a)–(c) From left to right, example images of plane, storage tank, and ship in xView. (e)–(g) From left to right, example images of plane, storage tank, and ship in DOTA. For cross-detection bands adaptation from the UCAS-AOD dataset to the ISPRS Potsdam dataset. (d) Car in UCAS-AOD. (h) Car in ISPRS Potsdam.

pipeline of generating pseudo-labels and selecting training images is summarized in Algorithm 1, and the complete process of HES-ST is described in Algorithm 2.

IV. EXPERIMENTS

A. Datasets

We validated our proposed method in two different domain shift scenarios: 1) *cross-satellite/senor adaptation* and 2) *cross-detection bands adaptation*. The four remote sensing datasets used in our experiments are described in Table I, and the data considered as source and target domain are listed.

1) Cross-Satellite: xView → DOTA: Following the first benchmark for adaptive object detection in remote sensing established by Xu et al. [45], we constructed a domain adaptation scenario across different satellites. The xView [12] dataset is the source domain, which is captured by the WorldView-3 satellite. Unlike the xView dataset, the DOTA [13] dataset is acquired by various sensors and platforms, such as Google Earth, JL-1, and Gaofen-2 satellite, and contain muticlass objects with various scales, orientations, and shapes, which are closer to real remote sensing scenes. Thus, we use the DOTA dataset's publicly accessible training and validation sets as the target domain in our experiment. As shown in Fig. 6, DOTA provides a considerable shift from xView due to different sensors of satellites. In addition, the DOTA dataset is annotated with 15 common object classes at a coarse level. In contrast, the xView dataset provides a more fine-grained annotation of objects, including 60 child subclasses belonging to seven parent classes. Therefore, we merge the subclasses into the parent classes of xView to match with the category settings of the target-domain dataset DOTA [13]. For instance, we merge ten distinct ship child classes (e.g., maritime vessel, motorboat, and sailboat) into a single ship parent class. Finally, our experiments focused on the same three categories present in both the xView and DOTA datasets, i.e., plane, ship, and storage tank. Each image is cropped into the chips of size 1024 × 1024 pixels with an overlap of 200 pixels.

2) Cross-Detection Bands: UCAS-AOD → ISPRS Potsdam: As shown in Fig. 6, we noted the discrepancy caused by various multispectral detection bands in remote sensing scenes and established domain adaptation scenarios using

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RGB images to other types of multispectral images, such as CIR composites. The UCAS-AOD [79] dataset contains a vehicle dataset and an aircraft dataset selected from Google Earth aerial images. Some details of this dataset are provided in Table I. The vehicle dataset with 510 RGB optical images and 7114 annotated objects is used as the source domain data in our experiment. In contrast, the target-domain data consist of CIR images with different detection bands from the ISPRS Potsdam [80] dataset. This dataset comprises 38 fine-resolution image patches, with 24 patches from the training set and 14 patches from the validation set. These patches of the same size 6000 × 6000 pixels with a ground sampling distance of 5 cm are captured from the center area of Potsdam, Germany. Each patch includes a true orthophoto (TOP) along with corresponding digital surface models (DSMs), and TOP provides various channel compositions such as RGB, CIR, and all RGB and IR bands. In our experiment, we utilized only the CIR bands of TOP and sliced each patch into the chips of size 1024 × 1024 pixel with a 200-pixel overlap. We reannotated the patches according to the publicly accessible pixel-level annotations of the ISPRS Potsdam dataset to obtain instance-level annotations for the object detection task. Finally, we obtained a total of 2432 CIR image chips with 20 711 annotated cars from the ISPRS Potsdam dataset to serve as target-domain data. Note that the annotations are only available for performance evaluation during the test phase.

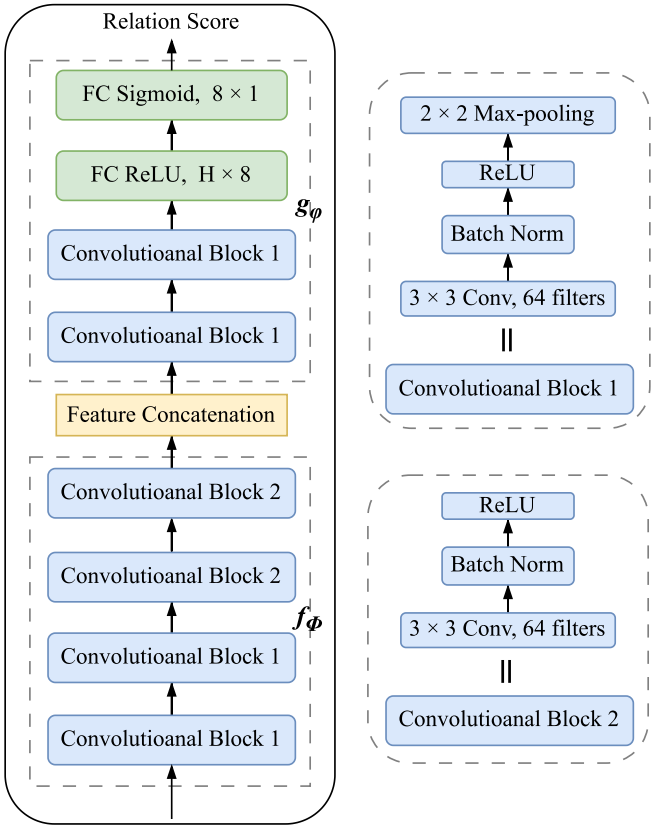


Fig. 7. Architectures of RN.

B. Implementation Details and Evaluation Metrics

For all experiments, we evaluated our methods using the standard object detection metric of average precision (AP) and its mean, i.e., mAP. We conducted each experiment on a GTX 2080Ti (11 GB) graphics processing unit (GPU).

1) YOLOv5 Detector Training Details: We employed the one-stage YOLOv5 [11] model as our base detector. YOLOv5 provides four kinds of networks (YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x), which have a similar network structure but differ in depth and width. For our experiments, we used the YOLOv5s model, which has the smallest network structure and the shallowest depth. The initialized network weights pretrained on ImageNet were used. We followed the original PyTorch implementation on hyperparameters of YOLOv5s unless specified. The input images were resized to 640 × 640 pixels, and IoU threshold for NMS was set to 0.45 in YOLOv5s. We trained the base detector YOLOv5s [11] on the source domain for 150 epochs with a batch size of 8 and a learning rate of 0.001. Fine-tuning, using the images obtained by GDT, the network was trained with a batch size of 8 and a learning rate of 0.001 for one epoch in cross-satellite adaptation and ten epochs in cross-detection bands adaptation. When fine-tuning the detector model with generated pseudo-labeled images, five epochs were performed in cross-satellite adaptation and one epoch in cross-detection bands adaptation for each HES-ST round.

2) CycleGAN Training Details: We trained CycleGAN with a learning rate of 0.0002 for the first 50 epochs and a linear decaying rate to 0 over the next 50 epochs. The input images were resized to 512 × 512 pixels due to the GPU memory limitations. The batch size was set to 1. For all other settings, we adopted the same hyperparameters and techniques as in the original CycleGAN article. For example, the least squares generative adversarial network (LSGAN) loss [51] was adopted for stabilizing our model training procedure. In addition, following the original article, we updated the discriminators of CycleGAN using a history of generated images rather than those produced by the latest generators to reduce model oscillation. The number of previously generated images stored in the image buffer was set to 50. After the CycleGAN was trained on the two domain images, we translated the source-domain images to the target domain through the learned mapping $G : D_s \rightarrow D_t$. In the GDT stage, we fine-tuned the YOLOv5 baseline detector using the target-style transferred images with source-domain annotations, and the specific training processes and details are described in this section under YOLOv5 detector training details.

3) RN Architectures and Training Details: The RN model adopted the same network architecture setting as the standard RNs [76], [78] designed for few-shot learning. As shown in Fig. 7, the embedding module contains four convolutional blocks. Each block has a 64-filter convolutional layer with a kernel size of 3, a batch normalization layer, and a rectified linear unit (ReLU) activation layer. Additionally, we apply 2 × 2 max-pooling in the first two blocks. The relation module consists of two convolutional blocks and two fully connected layers. Each of the convolutional blocks contains a 64-filter 3 × 3 convolutional layer followed by a batch normalization layer, a ReLU activation layer, and 2 × 2 max-pooling layer. The output size of the last max-pooling layer is set to $H = 64 \times 3 \times 3 = 576$. The first fully connected layer is 8-D with

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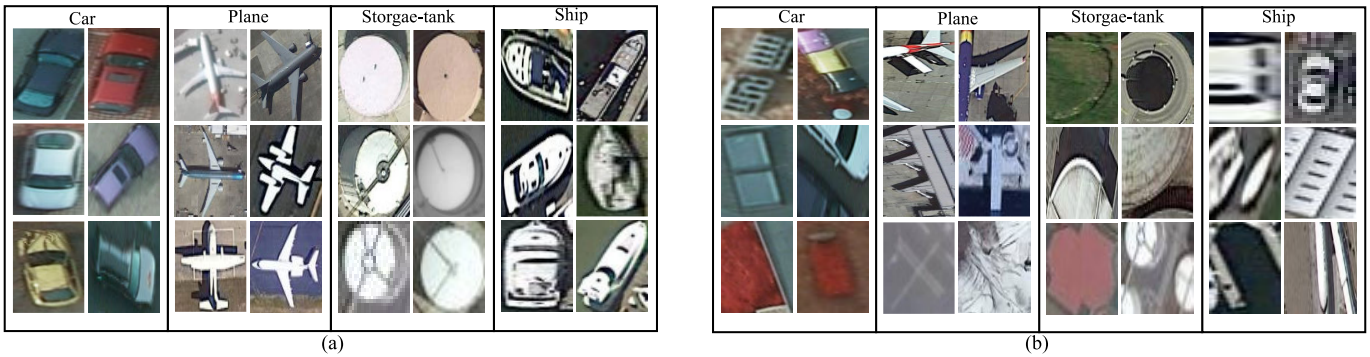


Fig. 8. Some collected training examples for RNs on each ST round. (a) Positive examples. (b) Negative examples.

ReLU, and the last fully connected layer is 1-D with Sigmoid in order to map the final relation scores into a reasonable range of 0–1.

For each object class of the target domain, we need to train an RN model from scratch with random initialization. Since we used the RN for binary classification with two-way one-shot training, we followed the standard training setup of the W-way K-shot adopted by the original RN article and most existing works. In each training episode, besides $K = 1$ support examples images, a total of $20 \times 2 = 40$ query examples were randomly sampled from the training example set with $W = 2$ categories. Note that we did not specially select an equivalent number of query images for each category as in [76]. In total, there are $20 \times 2 + 1 \times 2 = 42$ images in one training episode/minibatch. The input images were resized to 84×84 pixels. We used Adam optimizer with an initial learning rate of 0.001, which was annealed by half for every 100 000 episodes. For cross-satellite, each RN model was trained separately on a specific class of positive examples and 3252 negative examples from the source domain. Initially, there were 2886 positive examples for plane class, 9392 for ship class, and 3325 for storage-tank class. Then, in each HES-ST round, each model trained in the previous round was fine-tuned using updated positive and negative examples collected from the target domain for the corresponding category. Some collected examples are shown in Fig. 8. For cross-detection bands, the RN model is first trained on 7114 positive examples and 5125 negative examples obtained from the source domain. Then, in each subsequent HES-ST round, the model trained in the previous round was fine-tuned on updated positive and negative examples collected from the target domain for the corresponding category. Some collected examples are shown in Fig. 8.

4) Hyperparameters Setting: For the hyperparameters setting of the proposed algorithm, we mainly tune three hyperparameters in Algorithm 2, namely, θ in (1) of the DED, λ in (5), and ν of the CRJS. Here, θ determines the initial distance between the positive and negative decision boundaries. We set $\theta = 0.7$ for cross-detection bands and $\theta = 0.5$ for cross-satellite, respectively. λ controls the trust preference on detector or RN's prediction when calculating the CRJS. We would favor RN conservatively and set the weight λ to 0.3, as a default in all experiments. We argue that the outputs (i.e., object classes and bounding box locations) from the detector are simultaneously predicted in a highly coupled manner and sensitive to domain shift [37]. In contrast, the RN module is specialized in local image classification and reasonably deserves more emphasis when classifying examples. ν is the initial CRJS threshold for positive-hard examples selection; we set $\nu = 0.85$ for cross-detection bands and $\nu = 0.7$ for cross-satellite, respectively.

C. Experimental Results and Comparisons

We compare our method against the following ones.

1) *Baseline:* The base detector YOLOv5 was trained on the labeled source-domain data—xView for cross-satellite and UCAS-AOD for cross-detection bands. The trained baseline detector was then directly tested on the target-domain data—DOTA for cross-satellite and ISPRS Potsdam for cross-detection bands.
2) *Ideal Case:* We had access to the labels of the target domain in this case. The baseline detector model was fine-tuned directly using the target-domain data. The purpose of this experiment was to establish the upper bound of our proposed method.
3) *ST:* ST is the native ST method that uses a single confidence threshold to select above-threshold detections as reliable pseudo-labels. The generated pseudo-labels are then utilized to directly retrain the detector.
4) *PL:* Unlike ST and our method, which recreate pseudo-labels for every single epoch or ST round, PL in [18] generates pseudo-labels only once before training. The method of generating pseudo-labels in PL is identical to that employed in ST. For a fair comparison, ST, PL, and our method were all implemented using the baseline YOLOv5 [11] with the same training epoch setting. In addition, we set the threshold for both ST and PL to 0.5, which was their optimal parameter in our experiments.

1) Cross-Satellite: xView → DOTA: As given in Table II, compared to the baseline, applying GDT and HES-ST alone improves mAP by +8.6% and +8.0%, respectively. Moreover, the combination of GDT + HES-ST achieves 72.8% mAP and an improvement of 15.7% mAP compared to the baseline. The results indicate that both the GDT and HES-ST methods are effective and complementary to each other. On the other hand, the naive ST method showed a significant drop in performance with a −20.2% decrease in mAP compared to the baseline. This is due to the effects of false positives and false negatives in generated pseudo-labels (unreliable and incomplete).

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TABLE II
COMPARISON OF VARIOUS METHODS IN THE CROSS-SATELLITE ADAPTATION. THE DETECTOR IS TRANSFERRED FROM THE XVIEW TO THE DOTA DATASET

Method	Source	Target	Plane	Storage-tank	Ship	mAP
DAFaster(CVPR'18) [27]	✓		47.5	22.0	24.9	31.5
SWFaster(CVPR'19) [28]	✓		57.3	34.6	30.0	40.6
HTCN(CVPR'20) [31]	✓		54.7	34.6	30.0	39.3
EPMs(ECCV'20) [33]	✓		48.8	42.0	38.3	43.0
FADA(TGRS'21) [45]	✓		68.0	46.2	43.4	52.5
RetinaNet (TGRS'21) [8],[45]		✓	89.0	46.3	51.2	62.6
Baseline [11]	✓		65.3	35.2	70.8	57.1
ST [38]	✓		24.1	**65.0**	21.6	36.9
PL [18]	✓		64.9	40.5	75.8	60.4
Ours: GDT	✓		70.3	51.2	75.4	65.7
Ours: HES-ST	✓		72.5	49.6	73.3	65.1
Ours: GDT+HES-ST	✓		**80.0**	60.7	**77.8**	**72.8**
Ideal case	✓	✓	98.8	86.5	93.7	93.0

Moreover, this effect is accumulated/amplified during the ST process as we recreated pseudo-labels through the retrained detector each training epoch. The proposed HES-ST method outperforms the other two ST methods, ST and PL, by +28.2% and +4.7% mAP, respectively. We believe that HES-ST can generate high-quality pseudo-labels in each ST round. We will demonstrate this later in Section IV-D2.

Compared to prior SOTA works [27], [28], [31], [33], [45], our method achieves the current best performance listed in Table II. The work in [45] establishes this first benchmark (xView → DOTA) for domain-adaptive object detection in remote sensing, showing that the earlier SOTA work in standard computer vision based on the Faster R-CNN [7] detector, such as the representative works DA-Faster [27] and SW-Faster [28], achieves only 31.5% and 40.6% mAP, respectively. The current SOTA work HTCN [31] achieves 39.3% mAP, and the more recent work EMPs [33] based on the anchor-free one-stage FCOS [9] detector achieves 43.0% mAP. In remote sensing, the latest FADA [45] achieves a top accuracy of 52.5% mAP based on the RetinaNet [8] model and is the SOTA on this benchmark. Our approach, GDT + HES-ST, achieves a significant improvement of +20.3% mAP over the SOTA results for this benchmark. We emphasize that this performance even outperforms the upper bound method in [45] by +10.2% mAP. The upper bound method trained the base model RetinaNet [8] using target-domain labels in a fully supervised manner.

Due to the use of different detectors, our baseline, which trained YOLOv5s model on labeled source domain data, achieves a performance of 57.1% mAP, outperforming FADA [45] that based on the RetinaNet [8] with domain adaptation methods by +4.6% mAP. As reported in [45], this performance surpasses the source-trained RetinaNet (37.1% mAP) by +20% mAP and has only a −5.5% mAP performance gap compared to the target-trained RetinaNet (62.6% mAP). Theoretically, our CDST framework could be adapted to various detectors, allowing the flexibility to choose current high-performing and suitable detector, such as YOLOv5 [11] detector, for domain adaptation in remote sensing scenes.

2) Cross-Detection Bands: UCAS-AOD → ISPRS Potsdam: Table III shows the comparison of performance on cross-detection bands adaptation scenario using UCAS-AOD → ISPRS Potsdam. Compared to the baseline, using GDT and HES-ST alone improves AP by +14.5% and +10.9%, respectively. In contrast to cross-satellite adaptation, GDT achieves relatively larger improvement in cross-detection band adaptation. This is because the differences between the IR band and the other visible light channels are mainly reflected in the visual appearance differences of the images, resulting in a larger improvement potential for GDT. In addition, the smaller domain gap in the cross-detection bands (75.8 → 98.1, about 22.3% mAP gap) scenario relative to the cross-satellite (57.1 → 93.0, about 38.9% mAP gap) can make HES-ST work better and still outperform ST and PL by +12.2% and +4.5% AP, respectively. Applying both methods (GDT + HES-ST) results in an improvement of +16.8% AP compared to the baseline. We emphasize that this performance is only 5.5% lower than the Ideal Case in terms of AP and 3.6% lower in terms of precision.

TABLE III
COMPARISONS ON CROSS-DETECTION BANDS ADAPTATION USING THE UCAS-AOD AND THE ISPRS POTSDAM DATASET

Method	Precision	Recall	AP
Baseline [11]	83.9	69.5	75.8
ST [38]	86.4	71.1	74.5
PL [18]	87.0	75.7	82.2
Ours: GDT	90.4	85.1	90.3
Ours: HES-ST	87.6	81.4	86.7
Ours: GDT+HES-ST	**95.4**	84.5	**92.6**
Ideal case	99.0	96.7	98.1

D. Analysis of the Proposed Approaches

1) Ablation Study on HES-ST: To validate each component of the proposed HES-ST, we performed ablation studies in two adaptation scenarios: xView → DOTA and UCAS-AOD → ISPRS Potsdam. All experiments were conducted on the detector after implementing GDT step.

a) Effect and parameter sensitive analysis of CRJS: The parameter λ in (5) of CRJS controls the trust preference on the detector or RN's prediction when selecting hard examples. A larger λ value indicates higher trust in the

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TABLE IV
ABLATION STUDY ON HES-ST IN CROSS-SATELLITE ADAPTATION. PARAMETER SENSITIVITY ON CRJS IN HES-ST IS ANALYSIS AS WELL

xView → DOTA						
Method	CRJS/ λ	UE	Plane	Storage-tank	Ship	mAP
Baseline			65.3	35.2	70.8	57.1
HES-ST-1	0	✓	72.7	60.6	78.9	70.7
HES-ST-2	0.3	✓	80.0	60.7	77.8	**72.8**
HES-ST-3	0.5	✓	73.5	62.1	79.6	71.7
HES-ST-4	0.7	✓	77.7	58.2	77.3	71.1
HES-ST-5	1	✓	43.2	66.6	39.4	49.7
HES-ST-6	0.3		63.0	57.8	77.2	66.0

TABLE V
ABLATION STUDY AND PARAMETER SENSITIVITY ANALYSIS ON HES-ST IN CROSS-DETECTION BANDS ADAPTATION

UCAS-AOD → ISPRS Potsdam					
Method	CRJS/ λ	UE	Precision	Recall	AP
Baseline			83.9	69.5	75.8
HES-ST-1	0	✓	93.2	86.2	91.9
HES-ST-2	0.3	✓	95.4	84.5	**92.6**
HES-ST-3	0.5	✓	94.5	84.2	92.1
HES-ST-4	0.7	✓	93.7	83.7	91.1
HES-ST-5	1	✓	89.6	84.9	88.4
HES-ST-6	0.3		90.9	86.6	89.6

detector. Set λ to 1 disables CRJS (w/o CRJS), which means both easy and hard examples are selected only according to the detector's confidence level without RN. This is similar to the naive ST method but uses the CRJS threshold ν as the confidence threshold. According to the results shown as HES-ST-5 in Tables IV and V, this method degenerates the detection performance and only achieves 49.7% mAP for cross-satellite and 88.4% AP for cross-detection bands. On the other hand, incorporating CRJS, which combines detector with RN, provides a consistent gain over using only detector as seen in HES-ST-2–HES-ST-4 methods in Tables IV and V. Varying the weight λ can result in some fluctuation in performance, for cross-satellite, $\text{mAP}_{\lambda=0.3}$ is 72.8, $\text{mAP}_{\lambda=0.5}$ is 71.7, and $\text{mAP}_{\lambda=0.7}$ is 71.1. For cross-detection bands, $\text{AP}_{\lambda=0.3}$ is 92.6, $\text{AP}_{\lambda=0.5}$ is 92.1, and $\text{AP}_{\lambda=0.7}$ is 91.1. Set $\lambda = 0$ to choose fully trust the supervision from RN achieves the competitive performance of 70.7% mAP for cross-satellite and 91.9% AP for cross-detection bands as seen in the HES-ST-1 method. Therefore, CRJS, and particularly the inclusion of RN, is effective and essential for HES-ST, leading to significant improvements in detection performance.

b) Effect of UE: Comparing HES-ST-2 with HES-ST-6 method, we observed that without using UE harms performance with a drop of −6.8% mAP (72.8 → 66.0) for cross-satellite and a decrease of −3.0% AP (92.6 → 89.6) for cross-detection bands. Since UE can reduce not only FNs but also true positives (TPs) by selecting training images, we experimentally confirmed that minimizing the utilization of images containing detection errors (FPs and FNs) is crucial even though less images are used for training.

2) Quantitative Analysis of Pseudo-Labeled Image Quality: To further explore the causes of HES-ST's effectiveness, we quantitatively analyzed the pseudo-labeled image quality. We utilize both xView → DOTA and UCAS-AOD → ISPRS Potsdam as our study cases. We focus on pseudo-labeled images selected from the first and last HES-ST rounds for analysis, in order to demonstrate the improvement in pseudo-labeled image quality.

Inspired by [52], similar to the analysis of detection boxes, we categorize the pseudo-labels into five main groups according to the overlap (IoU as a criterion) and class between the pseudo-label box and the GT label box. The five categories are as follows.

1) *Correct (Cor):* Correct class, IoU meets the VOC [54] criterion for localization (IoU > 0.5).
2) *Best (Bst):* Part of the correct boxes that meet the strong localization criterion (IoU > 0.9).
3) *Localization (Loc):* Correct class, but with the misaligned box (0.1 < IoU < 0.5).
4) *Other (Oth):* Confusion with other objects, wrong class, IoU > 0.1.
5) *Background (BG):* Confusion with background, IoU < 0.1 for any object.

Then, the Cor boxes can be considered as the TPs of pseudo-labels, and they are reliable and effective pseudo-label boxes due to their correct class and accurate localization. On the other hand, the Loc, Oth, and BG boxes are three different kinds of FPs, which are unreliable pseudo-labels.

1) *Pseudo-Label Accuracy (PLA):* Generally, we can simply evaluate the quality of pseudo-labels based on their overall accuracy following the standard precision [53], [54] metric, measured by the PLA: the ratio of the number of reliable/useful pseudo-labels to the total number of pseudo-labels. It can be calculated as follows:

$$\text{PLA} = \frac{N_{\text{TP}}}{N_{\text{PL}}} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FP}}} \tag{6}$$

where N_{PL} represents the total number of pseudo-labels, which includes both N_{TP} reliable TP pseudo-labels and N_{FP} unreliable FP pseudo-labels.

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TABLE VI
QUALITY ANALYSIS OF PSEUDO-LABELED IMAGES IN CROSS-SATELLITE ADAPTATION

Method		Data Statistics						Quality Evaluation			Performance
		Images	PL	TP	FP	GT	Bst $^{Bst/Cor}$	PLA	WPLA	ACOR	mAP
1st	w/ UE	3171	54885	51600	3285	95279	2990 $^{5.8\%}$	94.0	98.2	**57.6**	**69.4**
	wo/ UE	4154	58510	54797	3713	109170	3126 $^{5.7\%}$	93.7	97.9	53.5	68.8
	Imgs_hard	983	3625	3201	424	13891	136 $^{4.2\%}$	88.3	94.6	**23.0**	-
Last	w/ UE	4651	84871	78337	6534	108138	6618 $^{\mathbf{8.4\%}}$	92.4	96.4	**78.6**	**72.8**
	wo/ UE	4766	74871	69166	5705	111254	4704 $^{6.8\%}$	92.2	97.0	62.1	66.0
	Imgs_hard	239	348	274	74	2811	7 $^{2.6\%}$	78.7	85.3	**9.7**	-

TABLE VII
QUALITY ANALYSIS OF PSEUDO-LABELED IMAGES IN CROSS-DETECTION BANDS ADAPTATION

Method		Data Statistics						Quality Evaluation			Performance
		Images	PL	TP	FP	GT	Bst $^{Bst/Cor}$	PLA	WPLA	ACOR	AP
1st	w/ UE	1030	10856	10720	136	14928	544 $^{5.0\%}$	98.7	99.4	**71.8**	**91.0**
	wo/ UE	1801	14222	14040	182	20380	669 $^{4.7\%}$	98.7	99.4	68.8	90.3
	Imgs_hard	771	3366	3320	46	5452	125 $^{3.7\%}$	98.6	99.3	**61.7**	-
Last	w/ UE	1632	14648	14517	131	19667	824 $^{\mathbf{5.7\%}}$	99.1	99.5	**73.8**	**92.6**
	wo/ UE	1843	14806	14688	118	20500	737 $^{5.0\%}$	99.2	99.5	71.6	89.6
	Imgs_hard	219	428	417	11	824	22 $^{5.2\%}$	97.4	98.8	**50.6**	-

2) *Weak Pseudo-Label Accuracy (WPLA):* Considering that pseudo-labels with inaccurate localization ($0.1 < \text{IoU} < 0.5$) may bring smaller adverse effect than FNs, which refers to unlabeled objects. Thus, we also calculate the WPLA by expanding the range of TP from Cor to Cor + Loc and recounting N_{TP} in (6). However, we note that the PLA and WPLA metrics only focus on evaluating the reliability of the pseudo-labels and ignore their completeness in the image.
3) *Accurate Object Capture Ratio (AOCR):* To provide a more comprehensive quantitative analysis of the pseudo-label quality for object detection, we adapt the standard recall [54] evaluation metric to the AOCR, which is defined as the percentage of reliable pseudo-labels covering GT boxes. It can be calculated as follows:

$$\text{AOCR} = \frac{N_{\text{TP}}}{N_{\text{GT}}} = \frac{N_{\text{TP}}}{N_{\text{TP}} + N_{\text{FN}}} \tag{7}$$

where N_{GT} is the number of GT labels in all training images, which includes N_{TP} GT boxes that are accurately covered by pseudo-labels of TP, and N_{FN} missed GT boxes, denoted as FN. The AOCR evaluates the reliability by considering TP and reflects the completeness by accounting for FN in GT.

As mentioned previously in UE, for each HES-ST round, two types of reliable examples can choose three types of training images: images with only easy examples (Imgs_easy), images with joint easy and positive-hard examples (Imgs_joint), and images with only positive-hard examples (Imgs_hard). Utilizing all three types of images along with their corresponding pseudo-labels as pseudo-labeled images, denoted as wo/UE, is a rudimentary approach that does not employ an image selection strategy. On the other hand, our UE approach simply employs Imgs_easy and Imgs_joint, denoted as w/UE, and this excludes a group of images only contained hard examples, denoted as Imgs_hard. The results are shown in Tables VI and VII, and we have several interesting observations as follows.

a) UE provides the higher quality among the three main groups of pseudo-labeled image: The effectiveness of various methods can be directly compared by evaluating the quality of the pseudo-labeled images selected in the first round since each method starts with the same performance detector. For the first round, w/UE gives higher pseudo-labeled image quality of 57.6% AOCR for cross-satellite and 71.8% AOCR for cross detection bands. The AOCR of w/UE is +4.1% and +3% higher than that of w/o UE in two scenarios, respectively.

b) Imgs_hard are low-quality pseudo-labeled images: We quantitatively demonstrate that the pseudo-labels on high-risk Imgs_hard are inferior. For the first round, the pseudo-labels of Imgs_hard achieves lower quality of 23.0% AOCR for cross-satellite and 61.7% AOCR for cross detection bands. This represents a significant quality gap of −34.6% AOCR and −10.1% AOCR, respectively, when compared to w/UE, which includes Imgs_easy and Imgs_joint. In addition, Imgs_hard achieves comparable quality in terms of PLA and WPLA compared to w/UE. For cross detection bands, the quality gap in terms of PLA between them is only 0.1%. This suggests that CRJS in HES-ST guarantees the reliability of each pseudo-label box, leading to a higher PLA and WPLA. On the other hand, the low quality of pseudo-labels for Imgs_hard is mainly attributed to FNs, which leads to incomplete pseudo-labels and a lower value of AOCR. Compared to the first HES-ST round, the quality of pseudo-labels generated on the last round for Imgs_hard is even worse (23.0% → 9.7% AOCR for cross-satellite and 61.7% → 50.6% AOCR for cross detection bands) and the quantity is also decreasing (3625 → 348, 3366 → 428).

By excluding such low-quality pseudo-labeled images of Imgs_hard in each round, UE can effectively fine-tune and improve the model with higher quality pseudo-labeled images. As shown in "performance" of Tables VI and VII, w/UE

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consistently achieves higher detection performance than wo/UE due to higher quality pseudo-labeled images. Moreover, excluding Imgs_hard is a low-cost solution. Specifically, on the last round, only 274 and 417 TPs are not used for training in the two scenarios, respectively. However, by excluding Imgs_hard, a significant reduction of 2537 and 407 FNs is achieved.

c) Quantity and quality of pseudo-labeled images generated by UE are progressively improving: During the ST process, the detector can boost its performance and thus yield more accurate detections, which help to improve pseudo-label quantity and quality. Taking the cross-satellite scenario as an example, as the model performance progressively improves from the first to the last round, the number of PLs increases significantly (54 885 → 84 871), while the number of high-quality TP boxes also grows (51 600 → 78 337), thus maintaining a high PLA of 92.4%. We emphasize that the number/percentage of Bst boxes also increases (2990 → 6618, 5.8% → 8.4%), which indicates the better localization.

d) Model performance is steadily improved by using high-quality pseudo-labeled images: We start this experiment from the detector that first conducted the GDT step, achieving the base/lower bound performance of 65.7% mAP for cross satellite and 90.3% AP for cross-detection bands. As shown in the "performance" column, fine-tuning the detector on pseudo-labeled images selected by w/UE steadily improves performance as the detector can learn object features directly from accurate pseudo-labeled boxes. After the first round of HES-ST, the model exhibits a significant performance gain for cross-satellite (65.7 → 69.4% mAP) and a consistent gain for cross-detection bands (90.3% → 91.0% AP). Finally, the model achieves 72.8% mAP and 92.6% AP after the last HES-ST round in two adaptation scenarios, respectively.

e) Performance degradation or stagnation occurs due to the error accumulation: For cross-satellite, w/o UE achieves 68.8% mAP after the first round of HES-ST, showing an improvement of 3.1% mAP compared to the lower bound. However, the performance is degraded to 66.0% mAP after the last HES-ST round as the result of erroneous pseudo-labels, which iteratively reinforce themselves and mislead the model to learn incorrect features. This trend is also observed in the cross-detection bands scenario (90.3% → 90.3% → 89.6% AP). Moreover, in both adaptation scenarios, w/UE can help to maintain the trend of performance improvement. However, the model still stops to further improve its performance after three (last) rounds due to a large number of label errors, as indicated by the imperfect quality of 78.6% and 73.8% AOCR in two adaptation scenarios. Error accumulation has been observed and discussed in many recent works on ST [35], [81], and it remains an open issue. Moreover, it can be even more challenging for object detection since inaccurate classification and localization can both contribute to the label error. Exploring more effective solutions for error accumulation in object detection is our future work.

3) Detection Performance Analysis Focusing on Errors: To understand the types of detection errors that is reduced by our two-step framework, we analyze the detection trends with ranks based on the tool from [52]. As in Section IV-D2, the predicted bounding boxes are also divided into five groups:

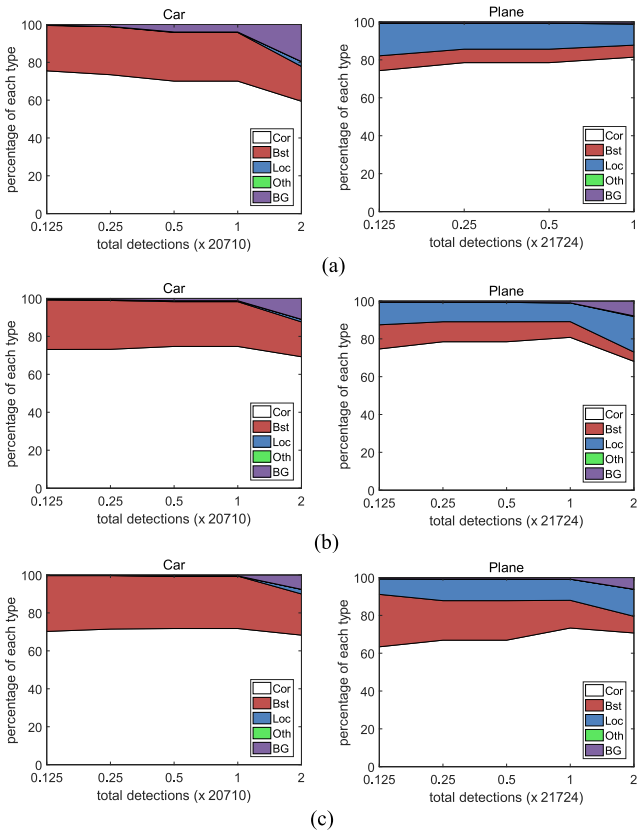


Fig. 9. Visualization of detection performance for different methods on car in ISPRS Potsdam dataset and plane in DOTA dataset. (a) Baseline. (b) GDT. (c) GDT + HES-ST.

two types of correct detections, Correct (Cor) and Best (Bst), and three types of error detections, Localization (Loc), Other (Oth), and Background (BG).

Fig. 9 shows the results of detection error analysis for "car" in the target domain of cross-detection bands (ISPRS Potsdam dataset) and "Plane" in the target domain of cross-satellite (DOTA dataset). Comparing the baseline and GDT, we observe that using GDT enhances the model's detection performance as the white- and red-color stacked areas, which represent the correct type of detections, are pushed upward, especially in high-confidence detections. For "Plane" in cross-satellite, more background error occurs in less-confident detections, which may be due to the coarse-grained target style of images and ambiguous object feature. Comparing the GDT and GDT + HES-ST, we observe the model shows a further improvement in performance, especially in terms of better localization, as seen in the increased red areas of Bst. Localization and background error (Loc and BG) are also reduced by HES-ST, which provides accurate and sufficient pseudo-labels in the target domain to help the detector remove such errors.

4) Training Time Efficiency of the Proposed Approaches: Table VIII shows the training time efficiency of different modules in the proposed approaches. The proposed framework includes three modules that need to be trained, i.e., YOLOv5s detector, CycleGAN, and RN. For YOLOV5s detector training, fine-tuning at the GDT stage for the cross-satellite adaptation (xView → DOTA) and the cross-detection bands adaptation (UCAS-AOD → ISPRS Potsdam) took an average of 2.1 and 1.12 min/epoch, respectively. When fine-tuning during each

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TABLE VIII
TRAINING TIME EFFICIENCY OF DIFFERENT MODULES IN OUR METHODS

Modules	Training Phase	Time Unit	xView → DOTA	UCAS-AOD → ISPRS Potsdam
YOLOv5s Detector	Fine-tuning (GDT)	Minutes per epoch	2.10	1.12
	Fine-tuning (HES-ST)	Minutes per epoch	1.02	0.66
CycleGAN	Base Training	Hours per epoch	2.77	1.71
Relation Network	Episode-based Training	Seconds per episode	0.33	0.33



Fig. 10. Example images generated by (middle) GDT from UCAS-AOD to ISPRS Potsdam. The source images in (left) UCAS-AOD domain are transformed to match with the target images in (right) ISPRS Potsdam domain.

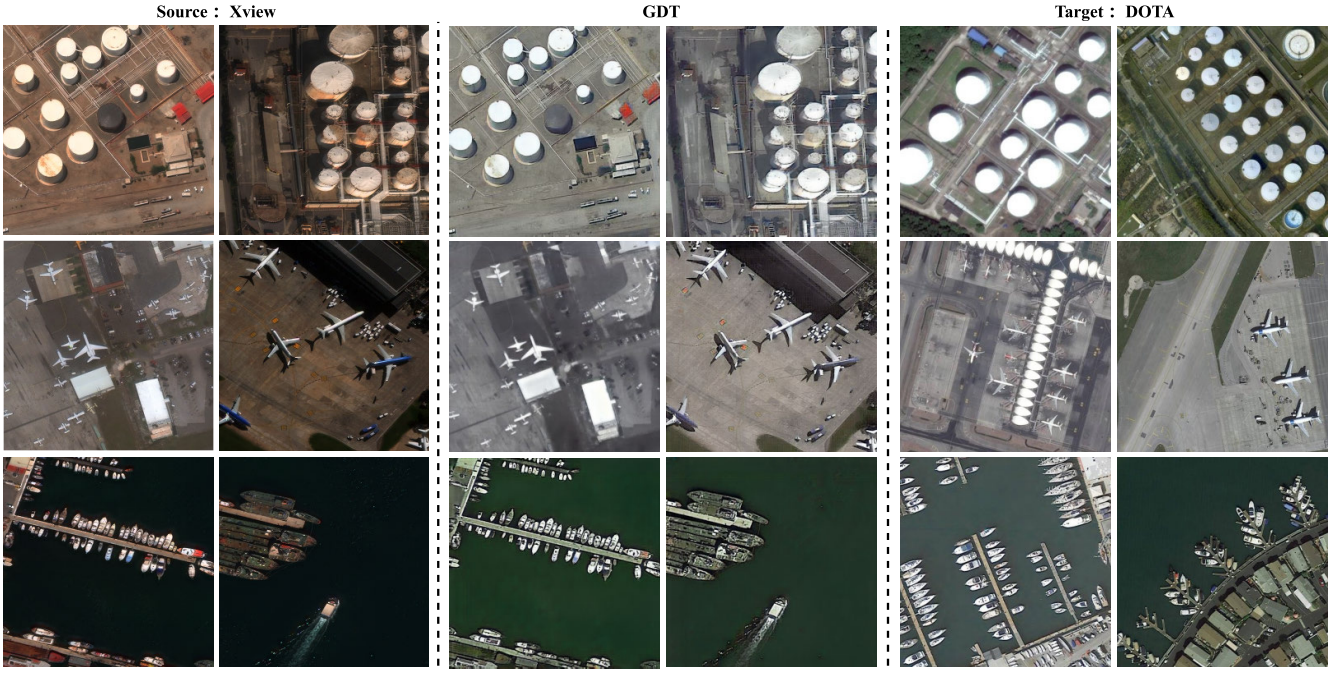
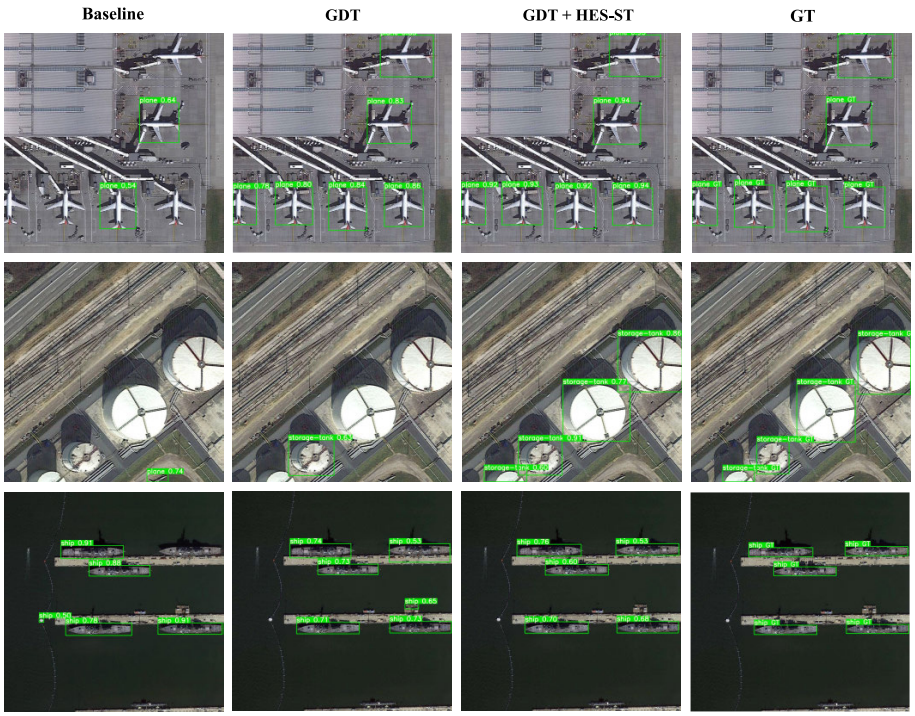


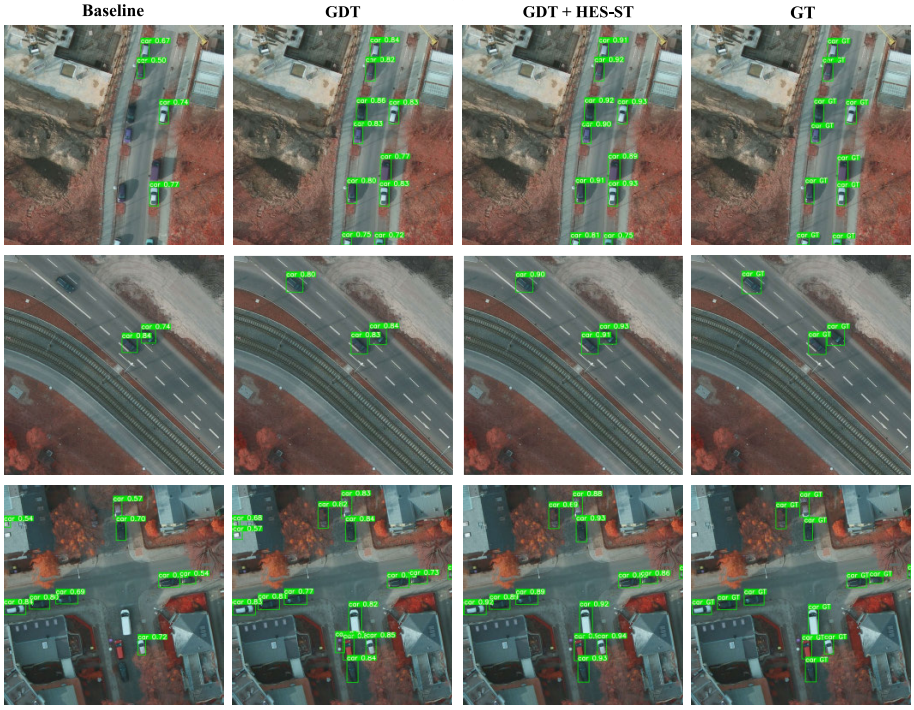
Fig. 11. Example images generated by (middle) GDT from xView to DOTA. The source images in (left) xView domain are transformed to match with the target images in (right) DOTA domain.

HES-ST round, the cross-satellite adaptation required a total of 5.1 min for five epochs, averaging approximately 1.02 min per epoch. The cross-detection bands adaptation took 0.66 min (39.6 s) for one epoch. For CycleGAN training, on average, each training epoch for cross-satellite adaptation was completed in 2.77 h, while each epoch for cross-detection bands adaptation was completed in 1.71 h. For RN training, each episode took about 0.33 s on average across different datasets. It is worth noting that in our experiments with the two-way one-shot training setup for the RN, each episode consistently includes a fixed number of training images, i.e., 40 query images and two support images. Consequently, the training time per episode remains unchanged across various adaptation scenarios. Overall, the YOLOv5s takes the shortest total training time, followed by RN, and CycleGAN takes the most time due to its high computational effort.

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(a)



(b)

Fig. 12. Qualitative detection results on two domain adaptation scenarios. (a) Cross-satellite: xView → DOTA. Visualization of detection results in the DOTA dataset. (b) Cross-detection bands: UCAS-AOD → ISPRS Potsdam. Visualization of detection results in the ISPRS Potsdam dataset. We present the results of the baseline, our GDT, GDT + HES-ST, and GT from left to right.

5) Impacts of Different Types of Detectors: To demonstrate the possible impacts of different types of detectors on our proposed methods, we investigate our CDST framework on other detectors such as Faster R-CNN [7] and SSD300 [3], following [18]. As shown in Table IX, compared to the baseline, the combination of GDT and HES-ST improves detection performance across various detectors in the cross-detection bands scenario, with varying performance gains in terms of mAP. Specifically, YOLOV5s improves +16.8%, Faster R-CNN improves +14.2%, and SSD300 improves +14.3%. The results demonstrate the generality of our CDST framework across all baseline detectors, and we believe that the proposed HSE-ST method is designed to exploit the outputs of the detector and thus can be applied to various detectors. Additionally, we found that the performance gain is significant in YOLOv5s compared to Faster R-CNN and SSD300. We emphasize

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TABLE IX
RESULTS OF OUR METHODS ON DIFFERENT BASELINE DETECTORS IN CROSS-DETECTION BANDS ADAPTATION

Method	YOLOv5s	Faster R-CNN	SSD300
Baseline	75.8	60.5	66.5
GDT	90.3	67.9	74.3
HES-ST	86.7	68.5	75.3
GDT+HES-ST	**92.6**	**74.7**	**80.8**
Ideal case	98.1	77.6	90.3

that using YOLOv5s as the baseline detector yields the best performance with an mAP of 92.6, highlighting the advantages of choosing YOLOv5s as a generic detector. Furthermore, previous works have mainly investigated domain adaptation for two-stage detectors, such as Faster R-CNN [7]. Therefore, this article employs YOLOv5s as a generic detector to explore the domain adaptation for single-stage detectors, which has not been fully investigated in remote sensing community.

E. Qualitative Results

1) Visualization of GDT: Figs. 10 and 11 show some example images generated by GDT, which transfers source-domain images to the target style. Visually, the overall appearance of the images looks quite similar to the target domain, and the contents are also preserved rather well. The colors and textures are transferred by CycleGAN while most edges and structures of objects remain. As recent works have demonstrated the validity of similar domain transfer approaches in nature [22] or cartoon [18] scenes, our experimental results of GDT in Tables II and III further confirm the effectiveness of our GDT method for optical remote sensing images.

2) Visualization of Detection Results: We compare the qualitative detection results of the baseline, our GDT, GDT + HES-ST, and GT to demonstrate the detection performance improvements as in Fig. 12. For both adaptation scenarios, we found that the proposed GDT method helps detector to detect more objects correctly with less FN errors compared to the baseline method. Moreover, HES-ST could further help the detector to predict a greater number of more accurate boxes with higher confidence. This implies that our adapted detector learns better objects feature representation of the target domain, which is potentially benefited from both target-style transferred images in GDT and target-domain pseudo-labeled images in HES-ST.

V. CONCLUSION

In this article, we propose a CDST framework performing the two-step adaptation to address the UDA for object detection in remote sensing. We first perform GDT to mitigate appearance differences for the detector and then launch out our HES-ST. Our method is validated on two remote sensing adaptation scenarios, and the experimental results outperform the baseline and other comparison methods by a clear margin, thus demonstrating the effectiveness of our framework. Besides, we achieve a new SOTA result of 72.8% mAP for the first standard benchmark [45] of domain adaptive object detection in remote sensing, with an impressive improvement of 20.3% mAP over the previous SOTA results.

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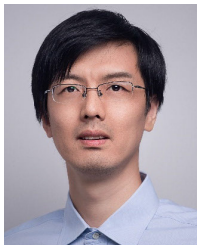
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